

Topological Sorting (Kahn's Algorithm / BFS)

→ linear ordering of
 vertices such that
 if there is an
 edge between
 u & v , u appears
 before v in the ordering

0 →

1 →

2 → {3}

3 → {1}

4 → {0, 1}

5 → {0, 2}

5 4 0 2 3 1

4 5 2 3 1 0

5	0
4	0



2.00



5 4 0 2 3 1
4 5 2 3 1 0

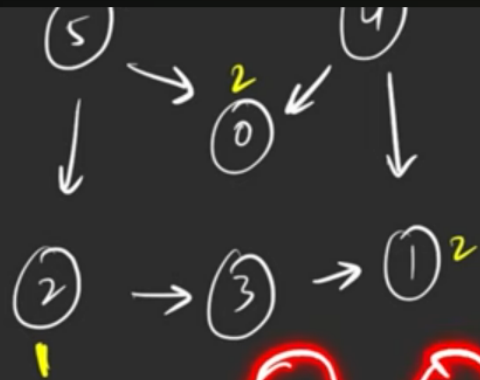
if there is an edge between u & v , u appears before v in the ordering

3 $\rightarrow$ {1}
4 $\rightarrow$ {0, 1}
5 $\rightarrow$ {0, 2}

5	0
4	0
5	2
4	1
2	3
3	1

6 edges





Count indegree

5
4

0

- 1 →
- 2 → {3}
- 3 → {1}
- 4 → {0, 1}
- 5 → {0, 2}

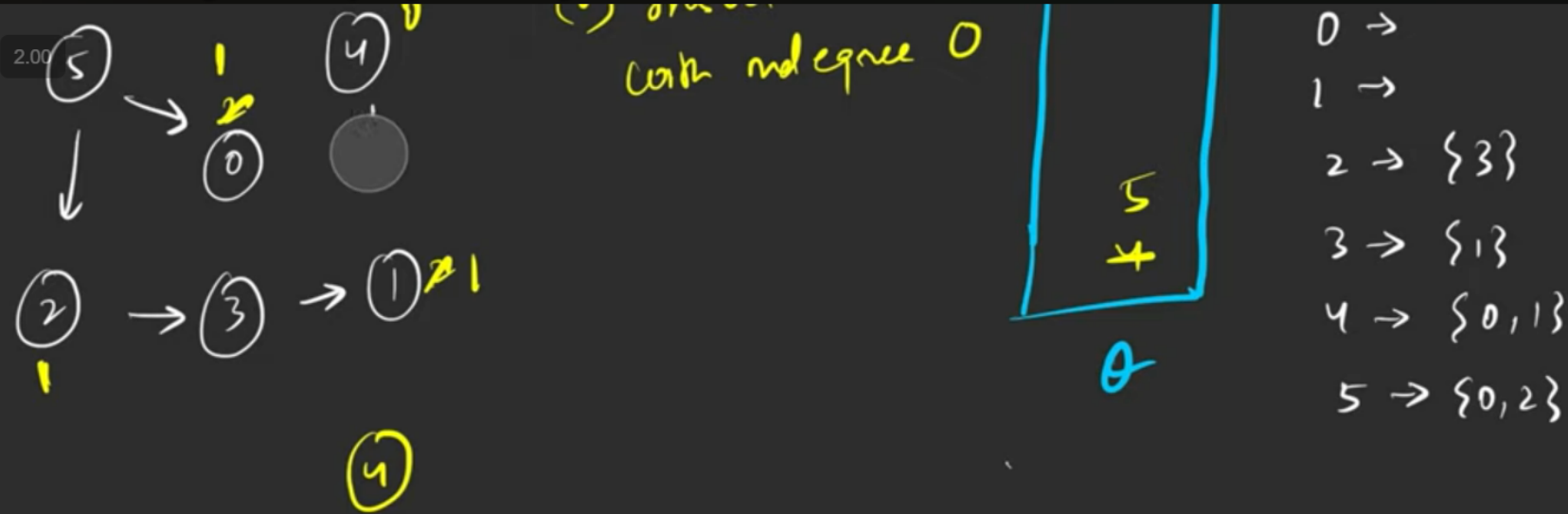


0	1	2	3	4	5
2	2	1	1	0	0

Indegree

→ no of incoming edges to a node





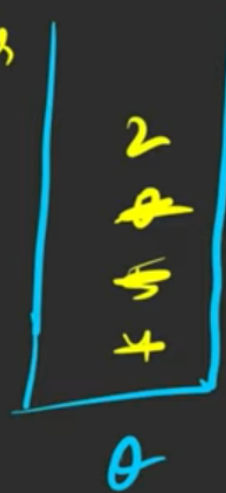
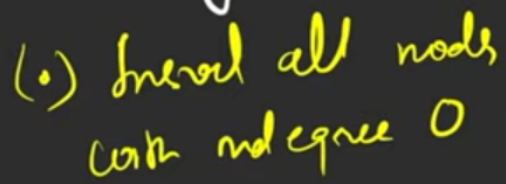
0	1	2	3	4	5
1	1	1	1	0	0

Indegree

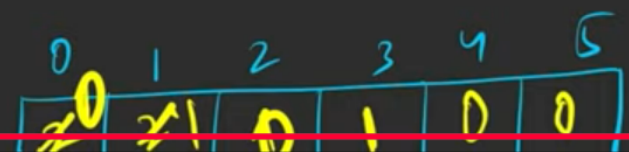
no of incoming edges to a node



Topology

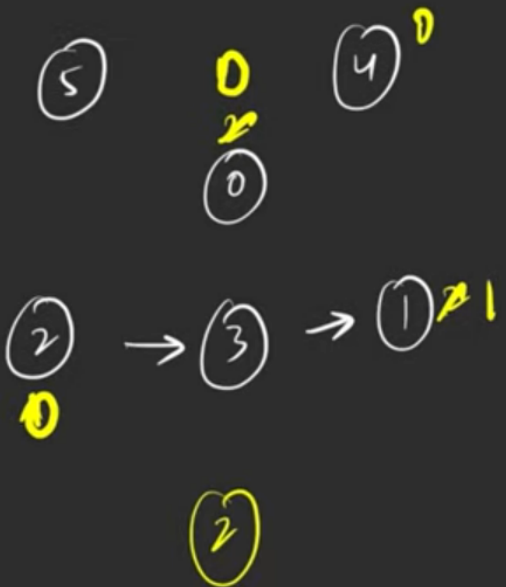

$$\begin{aligned} 0 &\rightarrow \\ 1 &\rightarrow \\ 2 &\rightarrow \{3\} \\ 3 &\rightarrow \{1\} \\ 4 &\rightarrow \{0, 1\} \\ 5 &\rightarrow \{0, 2\} \end{aligned}$$

450



Topological sorting

(.) Insert all nodes with indegree 0



- 3
- 2
- 4
- 5
- 1

- 0 →
- 1 →
- 2 → {3}
- 3 → {1}
- 4 → {0, 1}
- 5 → {0, 2}

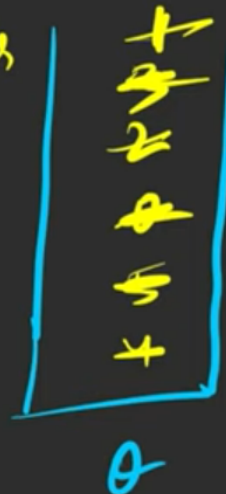
4 5 0 2

0	1	2	3	4	5
0	1	0	0	0	0

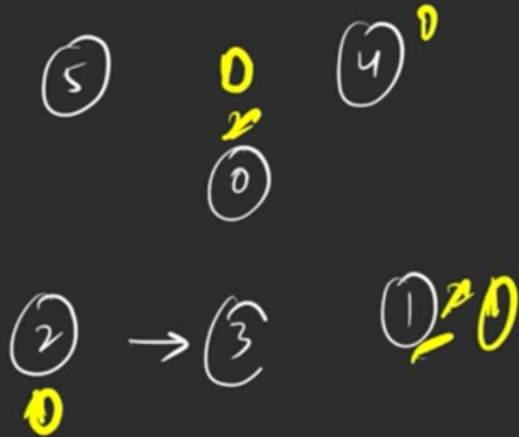


Topological Sorting (Kahn's Algorithm (BFS))

(.) Insert all nodes with indegree 0



- 0 →
- 1 →
- 2 → {3}
- 3 → {1}
- 4 → {0, 1}
- 5 → {0, 2}

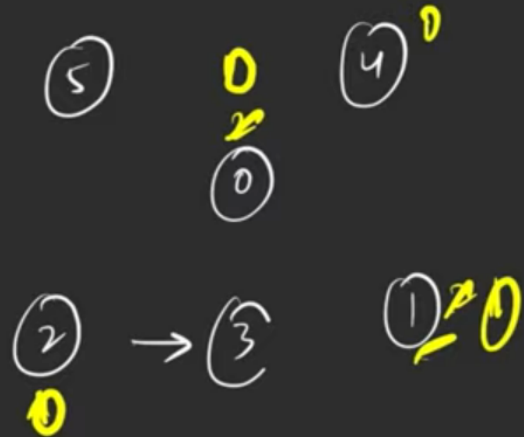


Topo Sort →

4 5 0 2 3 1

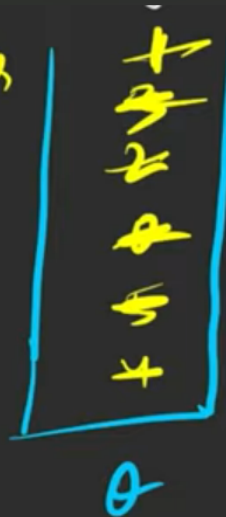


2.00



(.) Invert all nodes with indegree 0

(.) Take them out of the queue



0 →

1 →

2 → {3}

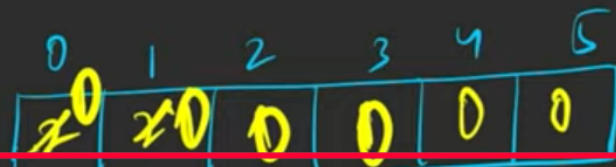
3 → {1}

4 → {0, 1}

5 → {0, 2}

Topo Sort ⇒

4 5 0 2 3 1



Settings



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C++ (g++ 5.4)

```
6 class Solution
7 {
8     public:
9         //Function to return list containing vertices in Topological order.
10        vector<int> topoSort(int V, vector<int> adj[])
11        {
12            int indegree[V] = {0};
13            for(int i = 0; i < V; i++) {
14                for(auto it: adj[i]) {
15                    indegree[it]++;
16                }
17            }
18            queue<int> q;
19            for(int i = 0; i < V; i++) {
20                if(indegree[i] == 0) {
21                    q.push(i);
22                }
23            }
24            vector<int> topo;
25            while(!q.empty()) {
26                int node = q.front();
27                q.pop();
28                topo.push_back(node);
29                // node is in your topo sort
30                // so please remove it from the indegree
31                for(auto it : adj[node]) {
32                    indegree[it]--;
33                    if(indegree[it] == 0) q.push(it);
34                }
35            }
36            return topo;
37        }
38    }
39    return topo;
```



Compile & Run



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